JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B6, PAGES 9873–9885, JUNE 10, 1991

PALEOMAGNETIC AND PETROGRAPHIC EVIDENCE FOR EARLY MAGNETIZATIONS IN SUCCESSIVE TERRA ROSA PALEOSOLS, LOWER PENNSYLVANIAN BLACK PRINCE LIMESTONE, ARIZONA

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Abstract. An early, synsedimentary, chemical remanent magnetization (CRM) is found in and below terra rosa paleosols which cap regressive carbonate cycles in the Pennsylvanian (Morrowan) Black Prince Limestone in southeastern Arizona. Authigenic hematite, occurring in fractures between soil nodules, in Liesegang bands, and replacing allochems, is abundant in specific horizons and is related to soil-forming processes. Paleomagnetic analysis of many specimens from the paleosol horizons reveals both northwesterly and shallow and southeasterly and shallow directions. These unidirectional, northwesterly and southeasterly magnetizations are interpreted to be antipodal equivalents and show a stratigraphic distribution with at least three reversals. The mean direction (declination 308°, inclination 9°) corresponds to a pole position at latitude 34°N, longitude 143°E similar to other pre-Kiaman, Carboniferous poles. Some specimens also contain multicomponent magnetizations (those with dual polarities or with curved thermal demagnetization trajectories) and are interpreted to be the result of remagnetization after a reversal in the Earth's field. Demagnetization and rock magnetic results for the reddened rocks confirm that hematite is the primary magnetic carrier. In contrast to the paleosol zones, the magnetization in the interbedded, gray marine limestones resides in magnetite and does not exhibit stable decay. Terra rosa paleosols are likely to have preserved the hematite and associated magnetizations that formed as a result of soil forming processes. These paleosols, therefore, may contain data that are useful in determining pole positions and validating polarity zonations.

Introduction

Paleomagnetic and magnetostratigraphic studies of Paleozoic units are commonly hindered because chemical and thermoviscous processes have remagnetized the rocks [e.g., McCabe and Elmore, 1989]. Because remagnetizations are so common, there is limited information on magnetostratigraphy and relatively few "high quality" reference poles from pre-Kiaman, Paleozoic units [e.g., Van der Voo, 1989]. Since these older rocks have a high probability of being remagnetized, it is particularly important to have independent indicators of primary remanence. Fold tests are one common and valuable constraining test, but can only provide a minimum age limit for a prefolding magnetization. Conglomerate and synsedimentary fold tests can also be used, but suitable rocks are rare, particularly in carbonate sequences. A stratigraphically controlled polarity zonation can also suggest a primary remanence, although the case for a primary remanence is strengthened if there are other tests (i.e., fold, conglomerate) which also support acquisition of an early magnetization.

Sedimentologic characteristics can also provide constraints on the timing of remanence acquisition. For example, magnetizations residing in hematite which formed during dedolomitization of ferroan dolomite [e.g., Elmore et al., 1985; Loucks and Elmore, 1986] must be secondary, following dolomitization. Other processes may bracket the time of magnetization in the depositional and diagenetic history of a sedimentary rock. One such process is the synsedimentary precipitation of hematite during formation of terra rosa soils. This study documents the presence of early magnetizations in paleosols in regressive carbonate sequences in the lower Pennsylvanian Black Prince Limestone in Arizona. These magnetizations reside in hematite that precipitated during soil formation and record reversals in the Pennsylvanian magnetic field.

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Paper number 91JB00338.
0148-0227/91/91JB-00338$05.00

Geologic Setting

The Black Prince Limestone in southeastern Arizona is biostratigraphically dated as early Pennsylvanian (Morrowan) and averages 50 m in thickness [Nations, 1963; Ross and Tyrrell, 1965; Ross, 1973]. The regional setting during Pennsylvanian time was a broad shelf where carbonates accumulated in shallow water (Figure 1). Goldhammer and Elmore [1984] describe five facies found in the Black Prince Limestone and relate them to depositional environments ranging from subtidal to supratidal or terrestrial. Calcrete or terra rosa soil facies cap regressive carbonate cycles that are composed of marine facies. Eleven paleosol-capped cycles were identified in the outcrops studied. Deposition and soil formation was presumably controlled by tectonic uplift or eustatic changes in sea level [Goldhammer and Elmore, 1984].

Subaerially exposed carbonates develop a range of features from karst to caliche depending mainly on climate, setting, and length of exposure [e.g., Esteban and Klappa, 1983]. Terra rosa soils contain abundant clay minerals and iron oxides that were added or concentrated during soil formation. Clays and iron oxides can be carried in by aeolian transport or preferentially concentrated as insoluble residues during weathering. This fine-grained material forms a red-brown matrix surrounding polyhedral, relict lithoclasts of limestone. The limestone lithoclasts are progressively less weathered in the deeper soil horizons until they merge into the limestone bedrock [Buol et al., 1980].

A list of observed features which are found at the top of the regressive sequences in the Black Prince Limestone and are indicative of the soil forming process include (1) red color, (2) nodules

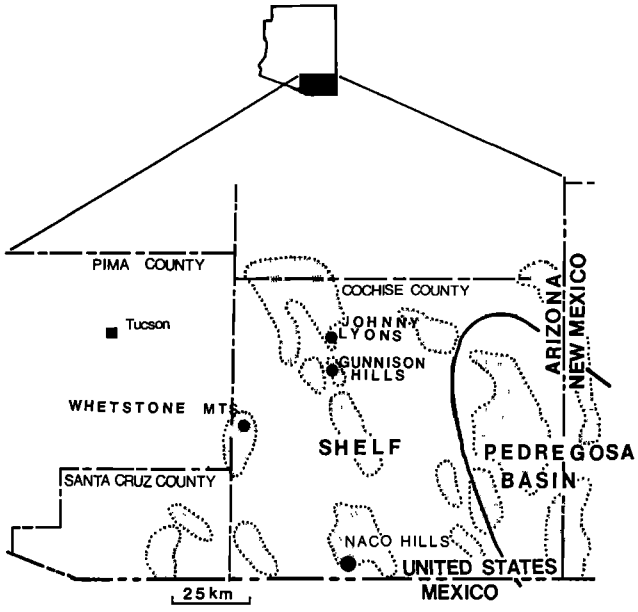


Fig. 1. Map of southeastern Arizona showing sampling locations and their regional relationship to the Pennsylvanian depositional setting. Sampling locations are in these sections: Johnny Lyons Hills, Sec. 16, T14S, R21E [Nations, 1963]; Gunnison Hills, Sec. 4, T16S, R33E [Nations, 1963]; Whetstone Mountains, Sec. 25, T18S, R19E [Ross and Tyrrell, 1965]; Naco Hills, Sec. 2, T2S, R23E [Goldhammer and Elmore, 1984].

produced by solution of parent limestone, (3) incomplete brecciation of the parent rock, (4) sharp nodule-matrix boundaries, with these boundaries truncating allochems and cements, (5) circumgranular cracking, (6) diffuse iron oxides within the nodules and in the carbonate below the paleosols, and (7) an insoluble matrix containing corroded allochems from the parent limestone, concentrated residual clays, and banded iron oxides [Goldhammer and Elmore, 1984].

The structural style of the study area in southeastern Arizona is the result of Triassic to Tertiary deformation characterized by normal faults and low angle reverse faults [Drewes, 1972]. From Triassic to Cretaceous time there was vertical movement of blocks across reactivated faults. During the Cordilleran orogeny (late Cretaceous to Paleocene), small igneous bodies were intruded into the Paleozoic section, and in subsequent tectonism there was thrusting of large sheets with horizontal movement of 10 to 100 km. Finally, during the Oligocene, the present Basin and Range configuration developed. The sampling sites for this study are located in hills uplifted by normal faulting. Rotation of structural blocks around a vertical axis has not been documented from the study area.

Conodont Alteration Index (CAI) values for southeastern Arizona range from high (6-7) near intrusions to lower values (<3) in areas only affected by burial [Wardlaw and Harris, 1984; Wardlaw et al., 1984]. The rocks near the Naco Hills section (Figure 1), where most samples were collected, have a CAI of 1.5-2 [Wardlaw and Harris, 1984]. Other sections sampled have similar values.

Sampling and Laboratory Techniques

The Black Prince Limestone was sampled at four locations (Figure 1). Within each section, terra rosa paleosols consist of reddish brown shale and/or reddened and brecciated limestone beds 1 to 50 cm in thickness that cap regressive cycles composed of gray marine limestones. Seven to ten cores (2.5 cm diameter) were drilled and oriented in the field from each site within the reddened limestones and above and below the shale units. Additional samples were taken from the gray limestones between paleosols for comparison. Hand samples and oriented slabs were also collected from some locations.

Standard specimens of 2.2 cm length were cut from the cores. The remanent magnetization in the specimens was measured with a United Scientific, three-axis cryogenic magnetometer located in a shielded laboratory. Stepwise, thermal and alternating field (AF) demagnetization was used to isolate and identify the components of the natural remanent magnetization (NRM). Specimens were heated at constant temperature steps (10-13 steps) for 30 minutes and then cooled to room temperature in a <2 nT field. The AF demagnetization was by external alternating fields of 10, 30, 50, 75, 90, 125, and 150 mT. Orthogonal projections [Zijderveld, 1967] of the magnetic directions and the intensity at each decay step were plotted for specimens and the principal component directions [Kirschvink, 1980] were calculated. A maximum angle of deviation (MAD) was calculated to assess the collinearity of the data. A MAD cut off of 6° was arbitrarily chosen. Remagnetization circle analysis [Halls, 1976] was performed on some specimens that exhibited curved demagnetization trajectories.

Coercivity spectrum analysis was used to aid in the identification of magnetic mineralogies [Dunlop, 1972; Lowrie and Heller, 1982]. Specimens were subjected in a stepwise fashion to an increasing isothermal remanent magnetization (IRM) of up to 0.8 T. Specimens were then thermally demagnetized at 13 temperature steps.

Chips from the drilled cores and hand samples from similar lithologies were used for the petrographic studies. Thin sections were examined in transmitted and reflected light and using cathodoluminescence microscopy. Fractured or polished and etched chips were examined with a scanning electron microscope (SEM) with energy dispersive analysis (EDA) capability.

Results

Paleomagnetism of Reddened Carbonates

Paleomagnetic results from reddened zones affected by paleoexposure indicate a complex magnetization history. Results from individual specimens were placed into groups based on the directional stability and directions of the magnetizations removed from each specimen (Table 1). About 30% of the samples have unstable magnetizations (e.g., MAD >6°) which invalidates principal component analysis, or curved demagnetization trajectories which could not be treated by remagnetization circle analysis. Three linear magnetic components were identified in the other specimens of the Black Prince Limestone. In addition, some specimens contain more than one

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TABLE 1. Total Number and Distribution of Components in Black Prince Limestone Specimens

Location	Total Analyzed	Component 1	Component 2	Component 3	MAD >6°	Remagnetization Circle Analyis
Naco Hills	144	62	33*	0	41	11
Whetstone Mountains	23	11	1	3	6	2‡
Gunnison Hills	97	9	3	35	50	0
Johnny Lyons Hills	40	0	18	12	10	0

Total specimens analyzed equals n_o in Table 2. MAD is maximum angle of deviation derived from principal component analysis [Kirschvink, 1980] and the number of specimens with MAD >6° includes those with weak and unstable magnetizations.

* Three of the 33 specimens also contained component 1 directions, but are only counted once for the total specimens analyzed.

‡ Specimens are suitable for remagnetization circle analysis, but the low number precludes statistical analysis. Strike and dip for the sampling locations are Naco Hills 177°E, 26°W; Whetstones 152.5°E, 17°W; Gunnison Hills 293.5°E, 38°NE; and Johnny Lyons Hills 322.5°E, 50°NE.

component, but the individual components could be resolved by principal component or remagnetization circle analysis.

Component 1. The Naco Hills, Gunnison Hills, and Whetstone Mountains sampling locations each had specimens with characteristic southeast and shallow or northwest and shallow magnetic directions which are referred to here as component 1. The magnetizations in these specimens decay in linear segments (Figures 2a, 2b) during progressive thermal demagnetization to moderate to high temperatures (300-680°C). Alternating field demagnetization of these specimens did not remove a significant amount of the NRM. Principal components calculated from demagnetization directions of the Naco Hills specimens, are southeast and shallow or northwest and shallow (Figure 3a; Table 2) after tilt correction. Principal component directions that were isolated in specimens from Gunnison Hills and Whetstone Mountains sections, however, were all southeast and shallow (Figures 3b, 3c; Table 2).

Beds are uniformily dipping at each location so no fold test is possible at a single location. A "regional" fold test was performed, however, between locations with different bedding orientations. Although the results of the test were not significant at the 95% confidence level (k1/k2=1.3, n=82), the results suggest acquisition of the remanence prior to tilting (Table 2). The mean magnetic direction from all sites, calculated after translating each southeast direction to its northwest antipodal equivalent, gives an overall mean direction of declination (Dec.)=310.9° and inclination (Inc.)=8.3° (Table 2).

At the Naco Hills location, good exposures made it possible to sample 11 successive terra rosa paleosols. The stratigraphic distribution of component 1 directions is summarized in Figure 4. Any one site predominantly contains either southeasterly or northwesterly directions, but almost every site also contains a few directions of the opposite polarity.

The demagnetization plots for some specimens reveal a dual polarity magnetization. For these specimens there is an abrupt change in the demagnetization trajectory at about 500°C from either northwest and moderately shallow to southeast and shallow, or southeast and moderate to northwest and shallow (Figure 5). The abrupt change in magnetization direction produces two linear components in single specimens. Principal component directions were calculated for both components.

Some specimens exhibited curved demagnetization trajectories in orthogonal projections (Figure 6a). When plotted as stereographic projections (Figure 6b), directions at individual demagnetization steps commonly fall on a great circle between the northwest and southeast and shallow directions, and during demagnetization the directions trended toward either the northwest or southeast. Great circle analysis [Halls, 1976] was applied to curved demagnetization trajectories from 11 specimens from the Naco Hills section. The distribution of the normals to the great circles and the resultant northwesterly-and-shallow point of least dispersion (Dec.= 308.0°, Inc.= 26.4°) are shown in Figure 7. This direction is somewhat steeper but generally consistent with the principal component directions obtained from linear components.

IRM acquisition curves show that the saturation magnetization is not reached by 0.7 T (Figure 8a) for specimens from the reddened to very reddened zones. During thermal demagnetization of IRM, maximum unblocking temperatures approach 680°C (Figure 8b). These results, and the demagnetization of NRMs, suggest that in the reddened carbonate zones, the only significant remanence carrier is hematite.

Component 2. In many samples a northerly and steep down component was removed during initial, low temperature (<300°C), demagnetization steps.

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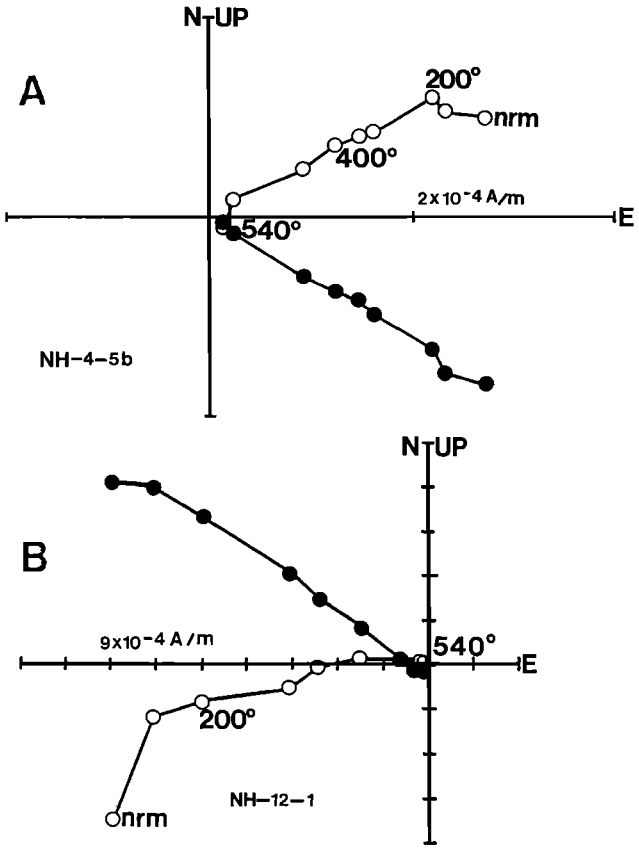


Fig. 2. Orthogonal projections of demagnetization data of representative specimens from the Naco Hills section. (a) Southeasterly and shallow magnetization direction removed during thermal decay trends toward the origin and is relatively linear. (b) Northwesterly and shallow magnetization direction removed during thermal decay of specimen also trends toward the origin. Solid symbols are projections onto the horizontal plane; open symbols onto E-W vertical plane. These symbols will be used in all orthogonal plots in this paper. Tick marks on all axes represent 1 x 10^{-4} A/m. Figures 2a and 2b are corrected for bedding tilt.

In other samples the entire decay path was northerly and steep down (Figure 9a). These directions from the Naco Hills section are plotted with their mean (Dec.= 0.3°, Inc.= 48.3°) in Figure 9b and listed in Table 2. Rock magnetic results for specimens containing component 2 are similar to those with component 1, but there is a greater percentage decay at low unblocking temperatures. These results suggest that hematite carries most of the remanence in specimens with this component.

Component 3. The third component was particularly common in specimens from the Gunnison Hills and Johnny Lyons Hills. The magnetization in these specimens typically shows linear decay during thermal demagnetization (Figure 10a) and is characterized by steeply negative inclinations. The magnetization in most of the specimens from the Gunnison Hills sites is steep with negative inclinations, but with a wide dispersion in declination (Figure 10b). The directions are not consistent with a drilling induced magnetization. Rock magnetic results are similar to those from specimens containing component 1, suggesting that the remanence is carried by hematite.

Petrography of Reddened Carbonates

Where best developed, the terra rosa paleosol facies is dark brown to red and contains nodules of the parent carbonate surrounded and capped by laminated claystones [Goldhammer and Elmore, 1984]. Not all the paleosols are complete, however, and some lack the claystone or the reddened limestone.

The claystone matrix that surrounds the carbonate nodules is composed of clays, carbonate fragments, quartz and chert silt, hematite, and carbonate cement. Hematite is often concentrated in Liesegang bands between soil lithoclasts (Figure 11a). The bands in the claystone are deformed by allochems during compaction. Other hematite is concentrated along suture-like boundaries between individual soil nodules within the matrix (Figure 11b). The friable nature of these soil facies did not allow sampling for paleomagnetic analysis.

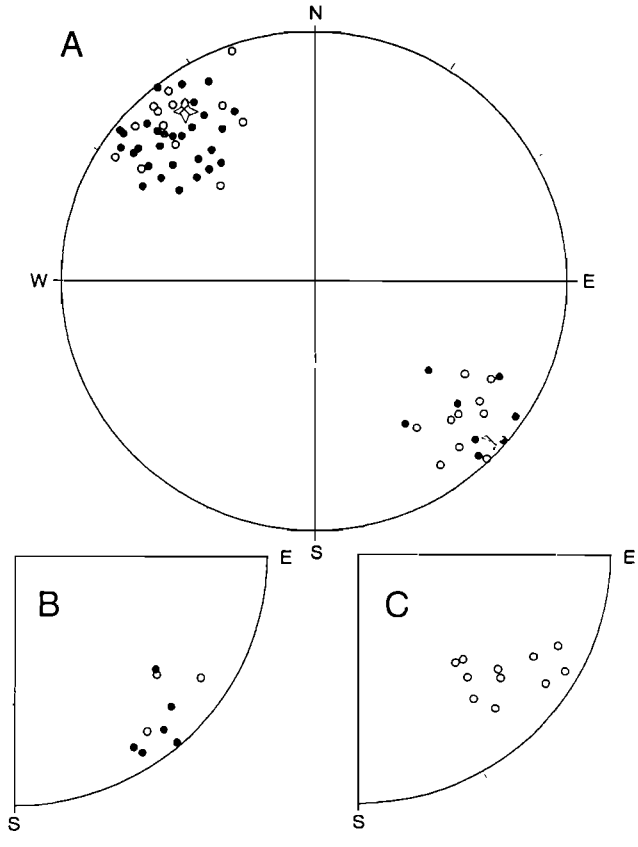


Fig. 3. Stereographic projections of principal component directions. (a) All component 1 directions from the Naco Hills sites. Means for the northwesterly and southeasterly groups are marked by stars. (b) All component 1 directions isolated from the Gunnison Hills sample sites. (c) All component 1 directions isolated from the Whetstone Mountains sites. Note that all of these sites are southeasterly. Open circles are projections on the upper hemisphere; solid circles, projections on the lower hemisphere (these symbols will be used in all stereographic projections in this paper).

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TABLE 2. Mean Magnetic Directions and Apparent Pole Positions for Components 1 and 2 From the Black Prince Limestone

Site	n/n_o	Dec. Angle	Inc. Angle	k	α95°	Lat.°N	Lon.°E	dp°	dm°
				Component 1					
Naco Hills									
Northwest,									
No tilt	44/144	320.2	23.0	24.1	5.9	48.6	141.3	4.6	8.6
Tilted		310.4	11.8			37.2	142.1	3.0	6.0
Southeast,									
No tilt	18/144	135.0	-5.6	6.8	14.8	38.9	135.4*	7.4	14.8
Tilted		133.5	-1.9			36.7	134.7*	4.4	8.8
Whetstone Mountains									
No tilt	11/23	135.9	-23.3	37.9	7.5	45.1	143.9*	4.3	8.0
Tilted		129.3	-20.5			38.8	146.9*	4.1	7.9
Gunnison Hills									
No tilt	9/97	129.2	-21.9	29.2	10.4	39.0	147.9*	5.8	10.4
Tilted		136.2	-1.4			38.1	131.6*	5.2	10.4
All Locations									
No tilt	82/264	314.0	19.0	9.7	5.6	42.3	143.5	3.0	5.8
Tilted		310.9	8.3	12.7	4.8	36.6	139.8	2.4	4.9
				Component 2					
Naco Hills									
No tilt	33/144	0.3	48.3	35.5	4.1	88.0	62.5	3.5	5.3

Dec. and Inc. are declination and inclination, respectively, in degrees; n/n_o is the ratio of specimens used in the statistical analysis to the number of specimens studied from each site (some specimens contain more than one component); k is the precision parameter associated with the specimen or site mean [Fisher, 1953]; α95 is the radius in degrees of the cone of 95% confidence of the mean; Lat. and Lon. are the respective latitude and longitude of the apparent pole; dp and dm are the semi-major and semi-minor axes in degrees, respectively, of the oval of 95% confidence around the poles associated with the paleomagnetic directions.

* The pole is the antipodal equivalent.

Beneath the clay-rich soil facies, the nodules grade into brecciated and reddened carbonates and finally into the gray to black marine limestones.

Most samples were collected from the brecciated and reddened carbonates. Chips cut from the ends of the core samples were used for petrographic analysis. Textures were compared directly to paleomagnetic results from individual specimens. The following hematite forms were seen in specimens containing component 1 and for specimens with multicomponent magnetizations.

One group of reddened carbonates contains hematite in Liesegang bands. The hematite is rhythmically concentrated in thin (1 mm scale) bands as individual plates or aggregates of specularite (Figure 12a). Some of these bands are cut by stylolites. These Liesegang banded dolomites and limestones may be comparable to Liesegang banded matrix material.

Calcite-filled fractures were often observed to have concentrations of iron oxides along their margins. Hematite completely fills some of the very thin fractures (Figure 12b). The fractures can be traced from the base to the top of the reddened carbonate zone and into the claystone matrix. Hematite also occurs along stylolites in the reddened carbonates. This hematite is much finer grained than other occurrences (Figure 12c).

Algal and crinoidal allochems in the reddened carbonates are cemented and partially replaced by specular hematite (Figure 13a). This type of hematite cement is commonly associated with or replaces pyrite. Hematite also occurs as individual dispersed crystals in very fine-grained, nonferroan dolomite and in limestones. The hematite is most abundant at intercrystalline boundaries, but is also found within individual dolomite grains. Many of the reddened limestones also contain dispersed,

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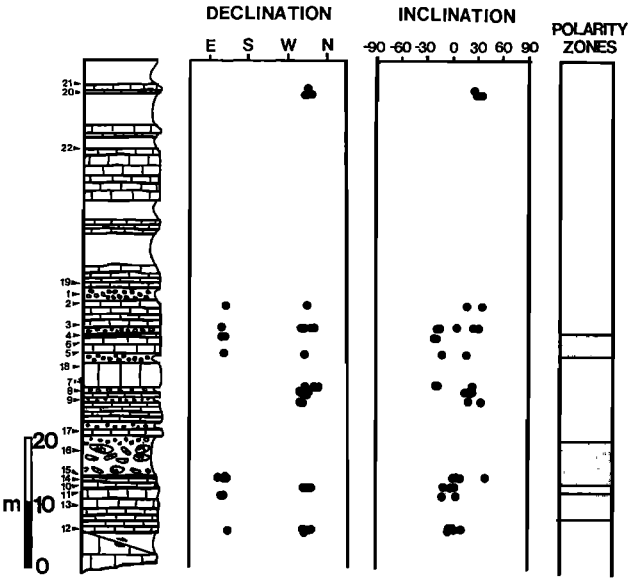


Fig. 4. Stratigraphic section showing sample sites, lithology, declination, inclination, and polarity zone interpretation from the Naco Hills section. Shaded portions of the lithologic column represent terra rosa paleosol horizons identified by Goldhammer and Elmore [1984]. Declination and inclinations are given for all samples showing linear, component 1 directions. Note that some sites contain both northwesterly and southeasterly magnetization directions.

individual or aggregate hematite grains (Figure 13b). Individual tabular crystals are up to 6 μm in diameter and occur in bladed, calcite cements and within allochems.

Gray Limestones

The magnetic characteristics of gray limestones, apparently unaffected by soil forming processes, were compared with the results from the interbedded, reddened carbonates. The AF and thermal demagnetization of many specimens from the

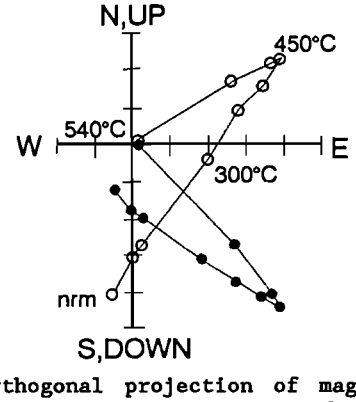


Fig. 5. Orthogonal projection of magnetization directions for demagnetization steps from a single specimen containing two magnetization directions. Thermal decay is not toward the origin in the initial demagnetization steps, but changes abruptly in direction and then decays toward the origin. Directions are corrected for structural tilt.

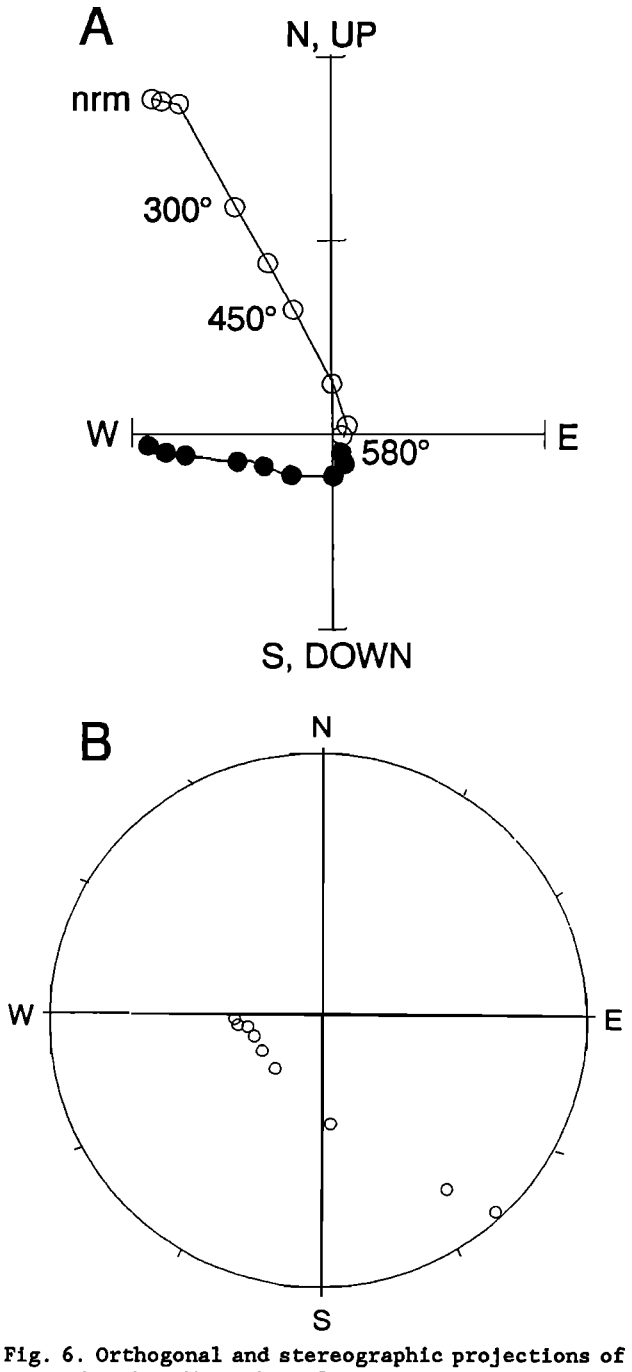


Fig. 6. Orthogonal and stereographic projections of magnetization directions for demagnetization steps from a single specimen (NH 3-4) containing two magnetization directions with overlapping blocking temperatures. (a) Trajectory of thermal decay is curved and is not towards the origin in the initial demagnetization steps. Directions are corrected for structural tilt. (b) The directions fall on a great circle on a stereographic projection and show a trend from a westerly to a southeast direction.

the gray limestones yielded erratic demagnetization trajectories which were at least partially due to weak NRM intensities. Other specimens showed removal of northerly and steep down components by low temperatures (200°C) or moderate field strengths (500 mT).

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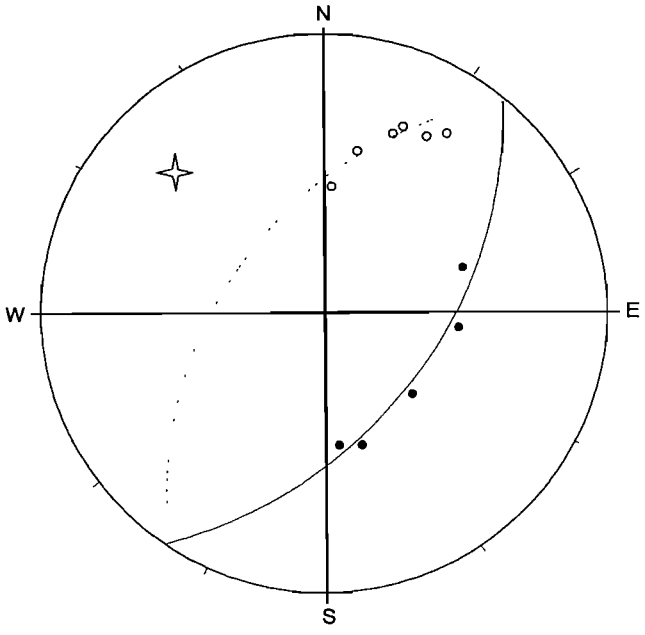


Fig. 7. Stereographic projection of data obtained from great circle analyses of samples with curved demagnetization trajectories from sites 7, 8, and 9 at the Naco Hills location. Solid circles are projections of poles to great circles on lower hemisphere; open circles are poles on upper hemisphere. Star is the pole to the plane defined by the poles from the individual specimens and represents a common direction (Dec.= 308.0°, Inc.= 26.4°).

Rock magnetic studies show that for gray and black limestones, saturation magnetization is almost reached by 0.2 to 0.8 T (Figure 14a). These same specimens have maximum unblocking temperatures near 580°C (Figure 14b), which suggests that magnetite is the magnetic carrier.

Interpretation

The paleomagnetic, rock magnetic, and petrographic results suggest that component 1 was acquired during the formation of the paleosols. Evidence supporting this interpretation includes the association between specimens with component 1 magnetizations and the paleosol horizons. Individual specimens containing component 1 magnetizations also contain authigenic types of hematite such as Liesegang bands, which precipitated as a result of soil forming processes. Northwest and southeast directions also have a stratigraphic distribution which suggest the remanence was acquired as the rocks accumulated during reversed and normal polarity intervals.

The stratigraphic distribution of specimens with opposite polarities suggests that the magnetizations can be grouped into magnetozones at the Naco Hills section (Figure 4). These magnetozones record reversals in the Earth's magnetic field that took place during the accumulation of the Black Prince and development of successive soil horizons. The Naco Hills section, therefore, passes a "reversal test", which suggests early acquisition of remanence. The results of the regional fold test, although not significant at the 95% level, are consistent with this interpretation.

Low CAI values of 1.5 to 2 [Wardlaw and Harris, 1984] and the stability of component 1 demagnetization directions to high temperatures (Figure 2) suggest that thermoviscous remagnetization did not affect these samples. Most sites contain either northwesterly or southeasterly directions. The predominant polarity at a single horizon is thought to reflect the polarity of the Earth's field during soil formation.

The presence of normal and reversed intervals in the Black Prince is consistent with polarity changes for the early Pennsylvanian reported by other workers [Khramov and Rodionov, 1981; DiVenere and Opdyke, 1988]. The pole position from the paleosols (Figure 15) is also similar to other Carboniferous poles reported for North America [e.g., Irving and Strong, 1984; Scotese et al., 1984; Kent and Opdyke, 1985; Murthy, 1985].

There is also evidence of partial remagnetization of these rocks. Although the magnetozones contain predominantly one polarity, some specimens within a given magnetozone are of an opposite polarity. There are also individual

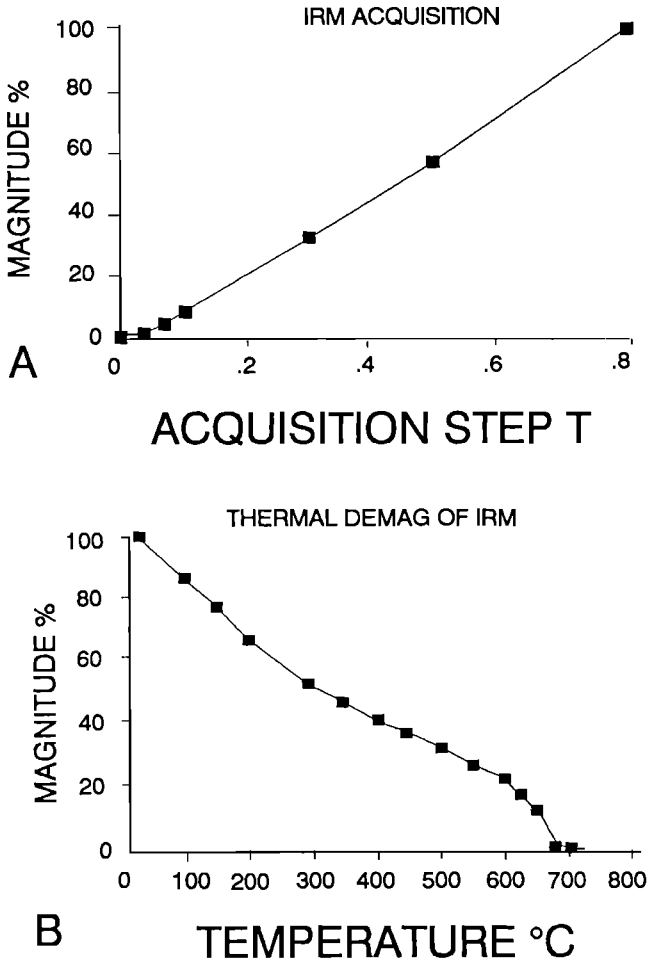


Fig. 8. Representative rock magnetic data from component 1 specimens. (a) IRM acquisition curve. Note gradual increase in acquired magnetization to 0.7 T indicating the presence of high coercivity minerals. (b) Thermal decay of partially saturated IRM show magnetization is not completely removed by 650°C, indicating the presence of hematite.

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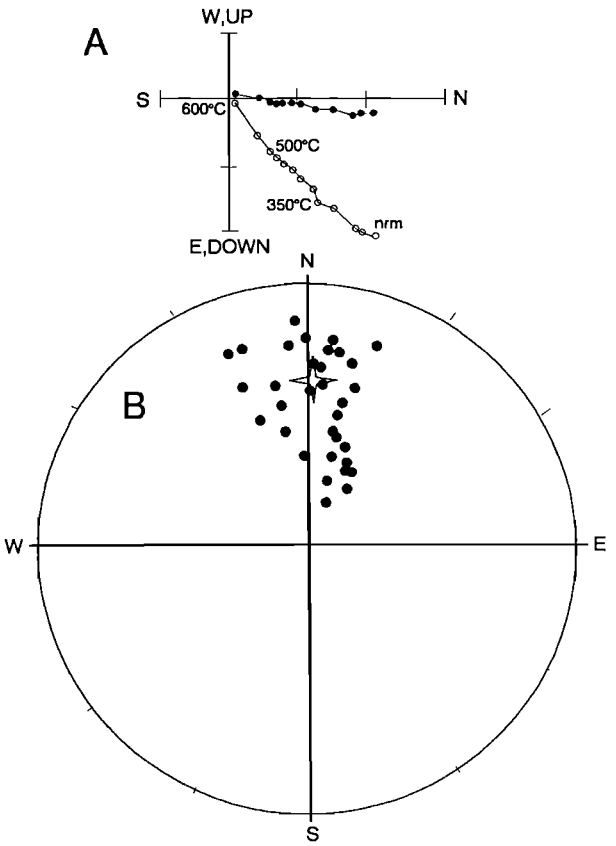


Fig. 9. Orthogonal and stereographic projections of magnetization directions in specimens with component 2 from the Naco Hills location. (a) Trajectory of thermal decay is generally linear and decays toward the origin. Directions are not corrected for structural tilt. (b) All component 2 directions are shown by solid circles and are projections on lower hemisphere. The mean direction is marked by a star.

specimens with multiple components, reflected in their dual polarities (Figure 5) or curved demagnetization trajectories (Figure 6). Acquisition of a multiple component magnetization is interpreted to be the result of a later, overprinting CRM acquired after a reversal in the Earth's field (e.g., Figure 5) or, possibly, acquisition through a reversal. Mechanisms that could produce secondary, overprinting CRMs are dehydration of goethite to hematite and/or precipitation of additional hematite in buried soil horizons due to fluctuating water tables. Based on petrographic analysis, specific types of hematite could not be associated with specimens that contain either reversed or normal directions. This is not surprising due to the complexity and length of time these units were affected by soil forming processes.

Component 2 directions are northerly and steep (Figure 9; Table 2) and the resultant pole position is latitude 85.5° N and longitude 171.8° E (Figure 15). This component is interpreted to reflect a modern CRM or viscous magnetization.

Interpretation of component 3 is problematic. The distribution of directions (Figure 10) is probably the result of a vector addition between components with similar blocking temperatures, but remagnetization circle analysis was not successful in isolating any dominant component. The predominance of negative inclinations suggests that acquisition of each component was in shallow or negatively inclined fields. The petrographic results suggest a CRM acquired long after deposition. These rocks contain abundant hematite along stylolites which may have been concentrated and altered during burial and pressure solution.

The gray marine limestones contain weak magnetizations, predominantly in magnetite, that are commonly viscous. There is no suggestion that they have retained a primary magnetization. Iron is mostly in the form of pyrite.

Discussion

One of the difficulties in refining the early to middle Paleozoic apparent polar wander path for North America has been a lack of units where suitable tests for primary remanence can be performed. Reports of primary remanent magnetizations in shallow water carbonates are not

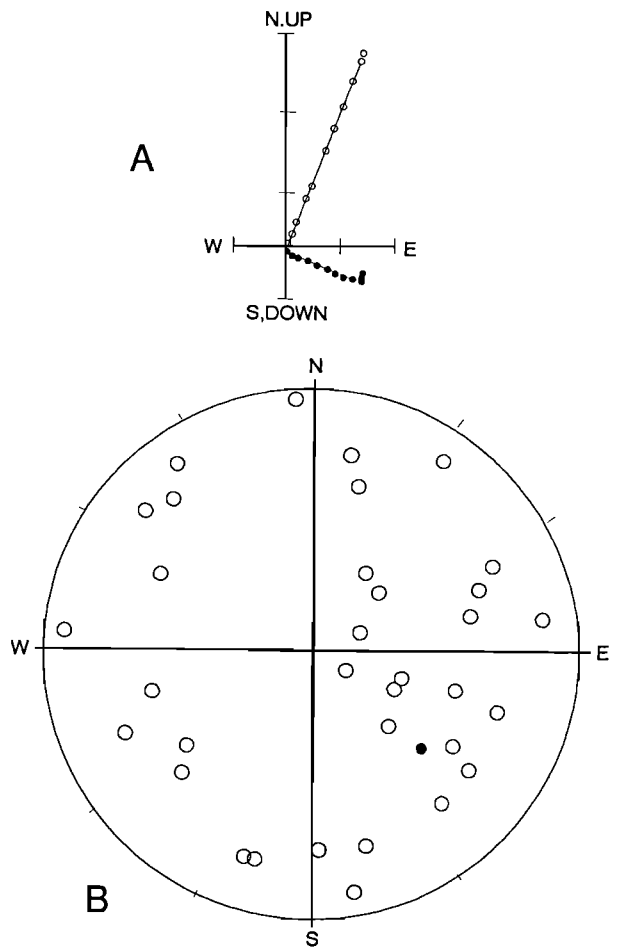


Fig. 10. Orthogonal and stereographic projections of magnetization directions in the component 3 group specimens from the Gunnnison Hills location. (a) Trajectory of thermal decay is linear and decays toward the origin. Directions are not corrected for structural tilt. (b) Component 3 directions.

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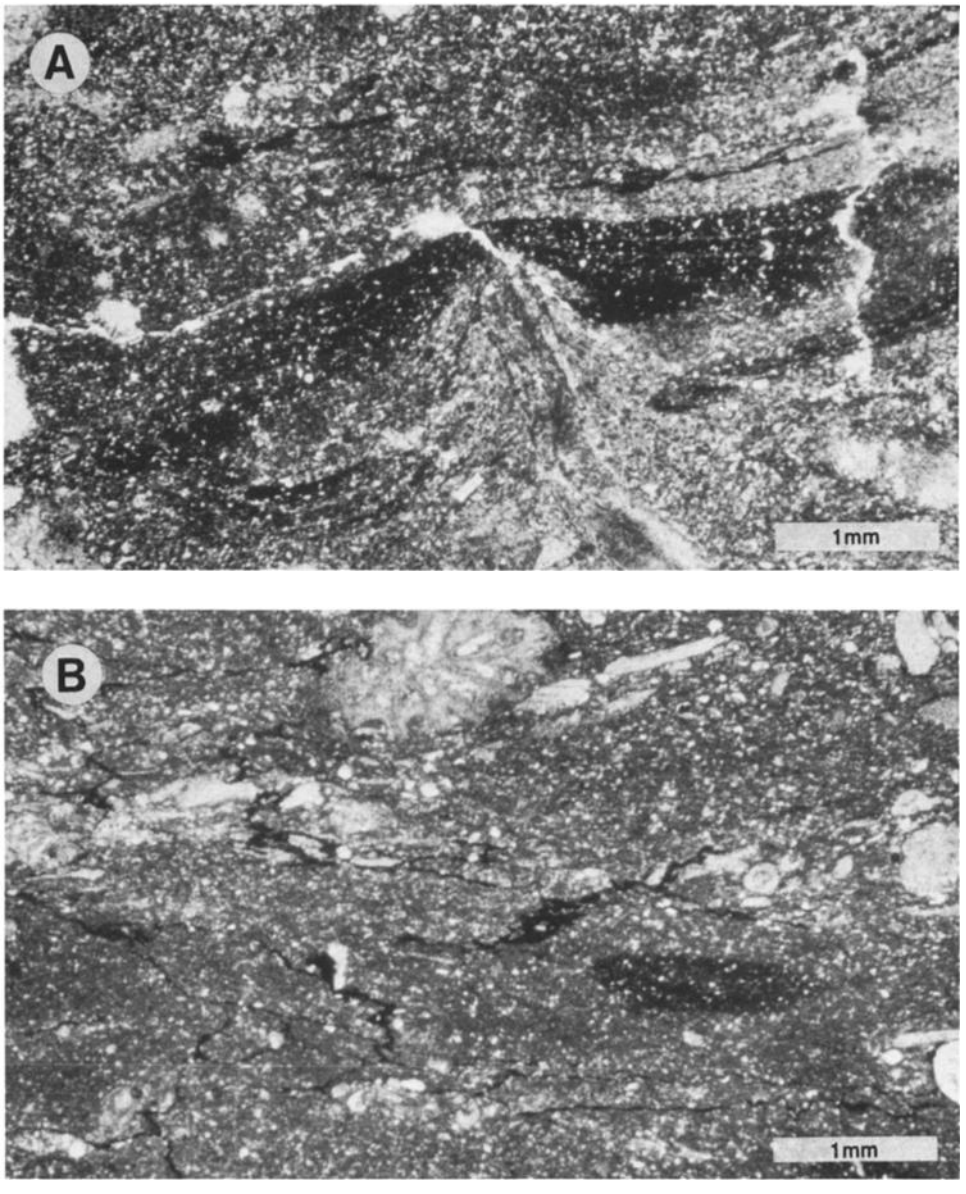


Fig. 11. Thin section photomicrographs illustrating hematite textures. (a) Liesegang bands in the clay-rich matrix material have been deformed during compaction. The deformation confirms early formation of hematite in the upper soil zone. (b) Sutured boundaries between soil glaebules contain concentrations of hematite (dark). The glaebules were formed during soil development.

common. Recently, however, there have been several reports suggesting or indicating a primary remanence in some carbonate units [e.g. McCabe et al., 1985; Hurley and Van der Voo, 1987; McNeill et al., 1988]. Hurley and Van der Voo [1987], for example, report the presence of layer-parallel polarity intervals in a condensed section of Devonian reefal limestones in Australia. They interpret the magnetization to be syndepositional, related to biogenic precipitation of iron hydroxides which altered to hematite during early diagenesis. McNeill et al. [1988] report a polarity zonation from late Neogene-Quaternary shallow-water carbonates. They report bacterial magnetofossils (magnetite) in the rocks, suggest the magnetization resides in this magnetite, and interpret the remanence to be primary or early diagenetic in origin. Although reversals in a sequence are usually used as evidence of a primary or early remanence, an apparent polarity sequence can also be caused by secondary components [e.g., Heller, 1978].

Five magnetostratigraphic zones were found in the Black Prince Limestone. The stratigraphic reoccurrence of paleosols in regressive carbonate cycles in general, increases the possibility that a "reversal" test can be performed for primary or early magnetizations.

A crucial problem in magnetostratigraphic studies is the completeness of the stratigraphic section [e.g., May et al., 1985; Tauxe et al., 1987]. This is particularly important for the Black

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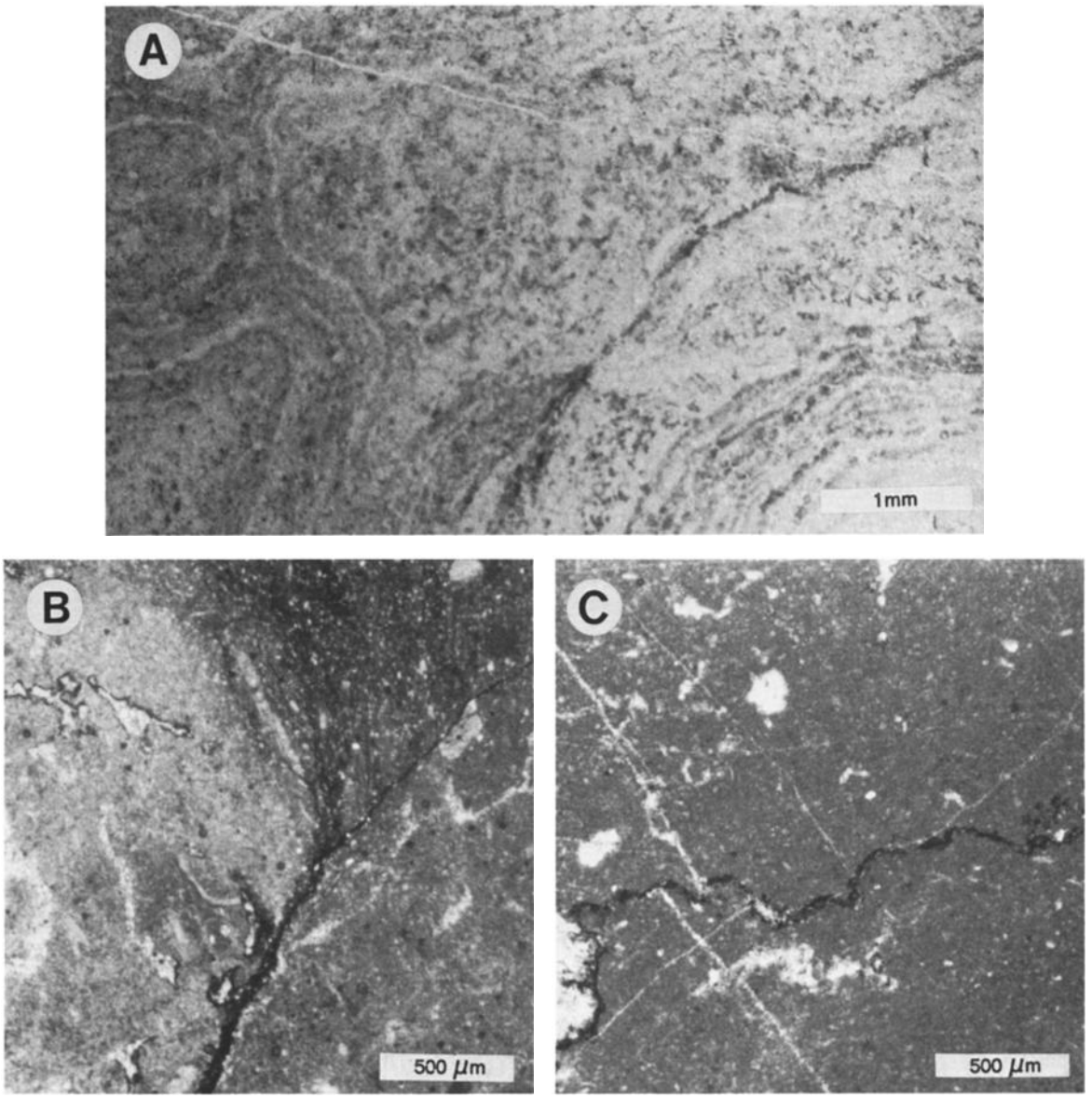


Fig. 12. Thin section photomicrographs of hematite in reddened carbonates associated with terra rosa soils. (a) Liesegang bands in dolomite probably formed in a similar manner to the Liesegang bands in the clay-rich matrix material. (b) Hematite filled fractures can be traced up into the overlying clay-rich and hematite-rich matrix material. (c) Hematite along stylolites is probably a late-formed texture.

Prince Limestone because the paleosols are obvious discontinuities where rock has been removed. Based on the approach of May et al. [1985], the probability that this Black Prince section records a polarity event of one million years duration is between 25 and 50%. Estimates of the probability, however, are complicated if only the paleosols are considered. It is clear that due to the incompleteness of the section it is likely that some reversals, particularly short duration events, will not be recorded. This indicates that caution should be used in correlating results from the Black Prince with other units. Even though magnetozones may be missing, the reversal stratigraphy is still useful, however, as evidence for an early or primary remanence.

There are additional complexities in CRM acquisition during periods of frequent magnetic reversals. For example, some specimens contain magnetizations interpreted to be of dual polarity. Acquisition of dual polarity magnetizations may be caused by the overprinting of a later, reversed, CRM onto a buried paleosol or by a continuation of soil forming processes through a reversal. Either process could produce a complex, multicomponent, NRM. Modern and Tertiary terra rosa paleosols could be studied to better understand the effects of reversals. Relatively long polarity intervals (e.g., Kiaman Reversed Superchron) may be required to enhance the preservation of a single component magnetization [e.g., Dunlop, 1986].

Because terra rosa paleosols are likely to

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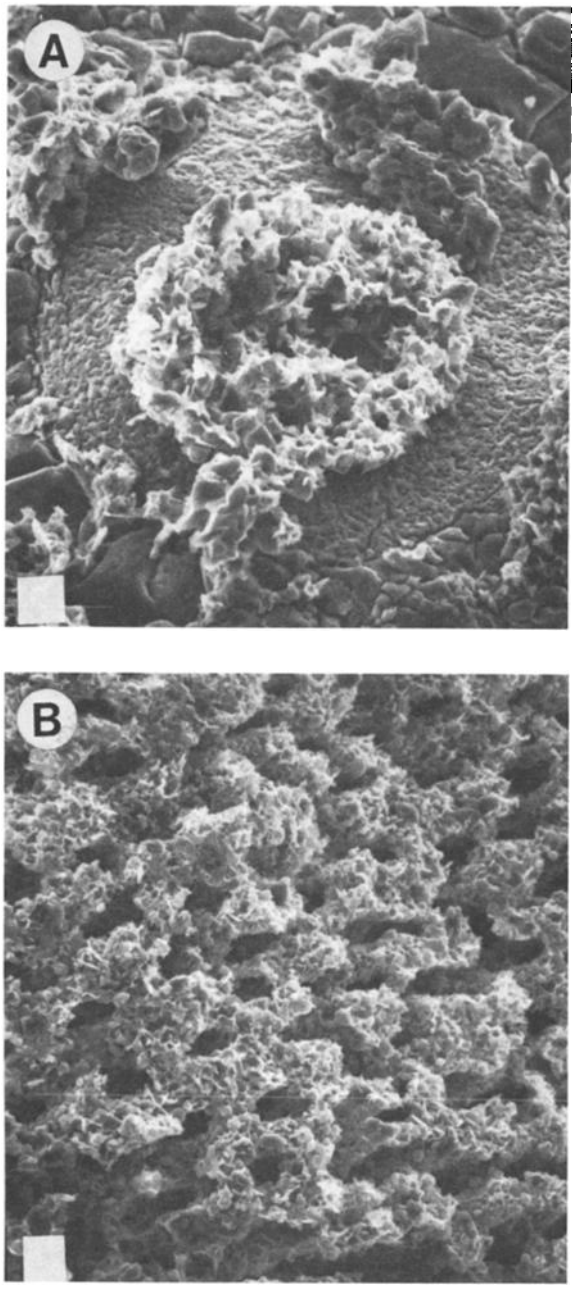


Fig. 13. Scanning electron micrographs of carbonates (acid etched)associated with terra rosa paleosols. The hematite is not dissolved and stands out in relief. Scale bars represent 10 μm. (a) The center of this crinoid columnal is filled with hematite. (b) Hematite (small flakes) is evenly dispersed in some limestones.

contain a primary magnetization, it might be possible to apply a soil test. As with other tests, identification of primary magnetizations by a soil test is not diagnostic in itself. A full assessment of paleomagnetic and petrographic factors should be made. Besides fold tests, terra rosa paleosols may often lend themselves to conglomerate tests. During soil development relict lithoclasts are isolated from the parent limestone. The blocks are affected by oxidative processes and can remain in place or be rotated [Goldhammer and Elmore, 1984]. A conglomerate test on disrupted blocks could constrain the time of acquisition. The Black Prince Limestone lithoclasts, however, were too small or too fractured to sample.

If the date of paleosol formation can be established with independent evidence, such as biostratigraphy, then the paleosols could be used to determine early or primary pole positions and perhaps polarity zonations. On the other hand, if a paleosol is not dated, it may be possible to use paleomagnetic analysis to constrain the time of formation of the paleosol. Another problem that could benefit from paleomagnetic analysis is the origin and timing of dolomitization in regressive carbonate sequences. Liesegang banded hematite that carries an early magnetization is found in dolomite. In such cases, dolomitization would probably have caused the hematite to be remagnetized so the age of dolomitization is constrained by the age of the included hematite.

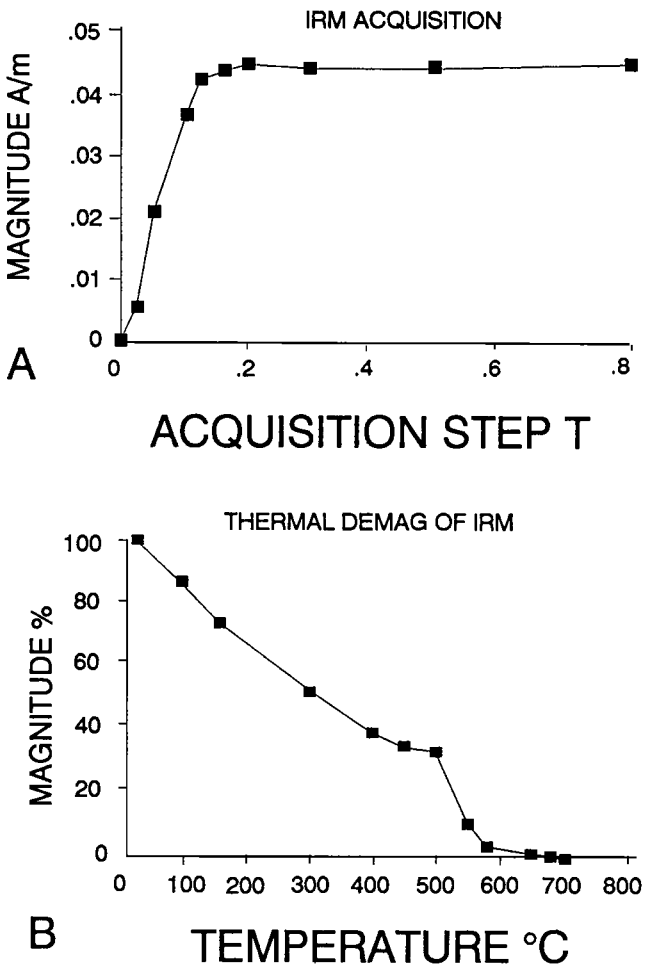


Fig. 14. Representative rock magnetic data from gray limestones. (a) IRM acquisition curve. Note abrupt increase in acquired magnetization to 0.15 T which suggest dominance of a low coercivity mineral. The slight increase above 0.15 T and continuing to 0.8 T suggests the presence of a high coercivity mineral. (b) Thermal decay of partially saturated IRM shows 99% decay by 580°C. A small amount of hematite may account for remaining magnetization above 580°C.

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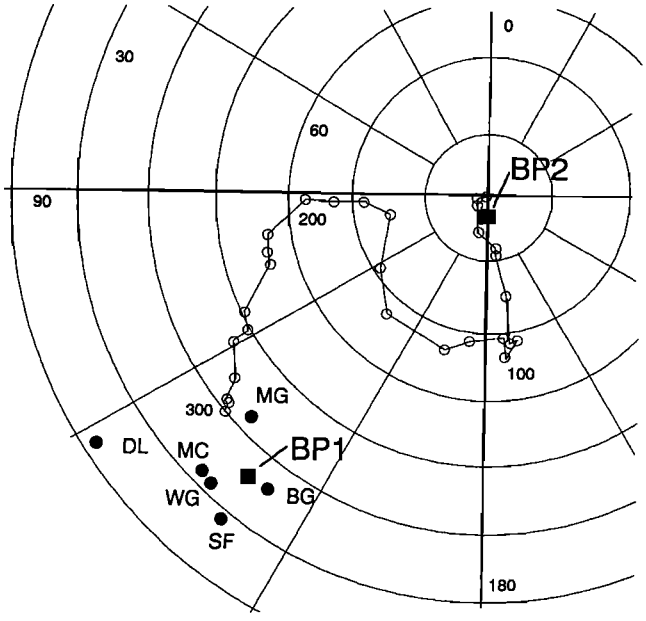


Fig. 15. Apparent polar wander path [after Irving and Irving, 1982] for North America with apparent magnetic poles from the Black Prince Limestone. BP1 is the pole calculated from component 1 magnetization directions. BP2 is the pole calculated from component 2 magnetization directions. Other Carboniferous poles are given for comparison: MG, Morien Group; WG, Windsor Group [Scotese et al., 1984]; BG, Barchois Group; SF, Spout Falls Formation [Murthy, 1985]; MC, Mauch Chunk Formation [Kent and Opdyke, 1985]; and DL, Deer Lake Formation [Irving and Strong, 1984].

Conclusions

The presence of an early magnetization in successive terra-rosa paleosols in the lower Pennsylvanian Black Prince Limestone is supported by paleomagnetic and petrographic data. Five polarity zones were identified in the Naco Hills section. Results of a "regional" fold test are also consistent with a prefolding magnetization, and the apparent pole position falls near other reported pre-Kiaman poles. Rock magnetic results suggest that the magnetic carrier in the soil zones is hematite.

The hematite in the soil zones is related to textures resulting from the soil forming process. These occurrences include hematite in Liesegang bands, as interparticle and intraparticle cements, in small fractures, and dispersed in carbonate cements. These rocks contain reversed and normal Pennsylvanian magnetizations.

A better understanding of the paleomagnetic and petrographic characteristics of terra-rosa soils could (1) make these focus points in the refinement of early to middle Paleozoic apparent polar wander paths and (2) illustrate how the palemagnetic method can be used to resolve sedimentologic problems.

Acknowledgments. The authors thank N. Donovon, Der Duen Sheu, M. Engel, R. DuBois, G. Atkinson and the anonymous reviewers for helpful comments on an earlier version of the manuscript.

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(Received July 12, 1990;
revised January 2, 1991;
accepted January 28, 1991.)

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